

A Finite Symbolic Mechanics Research Programme for Navier–Stokes Flow

From Continuum Smoothness to Finite Admissibility at an Alphonic Limit

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Research Monograph

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Abstract

The classical Navier–Stokes equations are among the most successful mathematical descriptions of fluid motion, yet the three-dimensional existence and smoothness problem remains unresolved. The conventional formulation assumes a continuum domain in which velocity and pressure fields can be differentiated at arbitrarily small spatial and temporal scales. Finite Symbolic Mechanics (FSM) proposes a different foundational starting point: every admissible representation is finite, every distinction has a lower representational bound, measurement carries uncertainty and provenance, and no physical or symbolic operation is permitted to rely upon an indefinitely refinable zero-extent point.

This monograph develops a research plan for applying FSM to incompressible Navier–Stokes flow without claiming to solve the classical Millennium Prize Problem. The intended study asks a different question. Rather than asking whether a smooth continuum solution must remain globally smooth, it asks whether a finite fluid trajectory can remain admissibly representable when spatial refinement is bounded below by an Alphonic limit, denoted α . The programme therefore constructs a finite graph- or cell-based fluid system in which mass, momentum, pressure relations, uncertainty and provenance are carried by finitely distinguishable regions and their interactions. Refinement is allowed only while the resulting regions remain no smaller than α . If the evolving dynamics would conventionally demand still finer distinction, the event is recorded as an Alphonic boundary event rather than silently replaced by further continuum refinement.

The monograph provides the conceptual background, a formal finite-state formulation, a staged computational protocol, Python-oriented pseudocode, a visualisation programme, nonlinear dynamical analyses, falsification and control tests, and a discussion of the possible interpretations of the results. The central objective is not to prove that continuum mathematics is false, nor to replace established computational fluid dynamics by assertion. It is to determine what remains of the Navier–Stokes difficulty when the continuum is treated as a highly effective mathematical compression of finite relations rather than as the foundational domain of physical description.

Contents

1	Introduction and Use Case	5
1.1	Why Navier–Stokes is an unusually useful FSM test	5
1.2	The immediate source of the question	6
1.3	The claim that is not being made	6
2	Historical and Mathematical Background	7
2.1	From continuum mechanics to Navier–Stokes	7
2.2	Weak solutions and the twentieth-century problem	7
2.3	The role of the continuum assumption	8
3	Finite Symbolic Mechanics Framework	9
3.1	Finite symbolic representation	9
3.2	The Alphonic limit	9
3.3	Finite time	10
3.4	Uncertainty and provenance	10
3.5	Admissibility	10
4	A Finite Graph Formulation of Fluid Motion	12
4.1	Why begin with a graph rather than a discretised derivative	12
4.2	Incidence operators	12
4.3	Mass balance	13
4.4	Momentum balance	13
4.5	Finite advection	13
4.6	Finite viscosity	14
4.7	Pressure as a finite constraint relation	14
4.8	Finite gradient and an Alphonic bound	14

5	The Alphonic Boundary Experiment	15
5.1	Defining a boundary event	15
5.2	An Alphonic boundary event is data	15
5.3	Three possible responses at the boundary	16
6	Research Questions, Hypotheses and Outcomes	17
6.1	Primary research question	17
6.2	Secondary questions	17
6.3	Working hypotheses	18
7	Experimental Programme	19
7.1	Stage 0: implementation verification	19
7.2	Stage 1: two-dimensional validation problems	19
7.3	Stage 2: three-dimensional Taylor–Green vortex	19
7.4	Stage 3: Alphonic refinement series	20
7.5	Stage 4: smooth high-shear initial-condition family	20
7.6	Stage 5: mesh and topology robustness	21
7.7	Stage 6: uncertainty study	21
8	Computational Architecture	22
8.1	Recommended software structure	22
8.2	Core data structures: Python-oriented pseudocode	22
8.3	Building an admissible graph	23
8.4	Finite transport and viscous exchange	24
8.5	Finite incompressibility projection	24
8.6	Refinement and Alphonic refusal	25
8.7	Full simulation loop	26
8.8	Matched classical-refinement control	27
9	Measurements and Diagnostics	28
9.1	Core quantities to record	28
9.2	Finite circulation rather than point vorticity	29
9.3	A boundedness audit	29

10 Visualisation Plan	30
10.1 Visualisation as part of the method	30
10.2 Figure set A: conceptual framing	30
10.3 Figure set B: flow fields	31
10.4 Figure set C: time-series plots	31
10.5 Figure set D: Alphonic scaling plots	31
10.6 Figure set E: trajectory visualisations	32
11 Nonlinear Dynamical Analysis	33
11.1 Why nonlinear analysis is useful here	33
11.2 Delay-coordinate reconstruction	33
11.3 Recurrence plots and recurrence quantification	34
11.4 Finite-time Lyapunov analysis	34
11.5 Correlation dimension and scaling cautions	34
11.6 Permutation entropy and symbolic complexity	34
11.7 Recurrence networks	35
11.8 Poincare sections and return maps	35
11.9 Transfer entropy as an optional spatial diagnostic	35
11.10 Linear comparison methods	35
12 Data, Provenance and Reproducibility	36
12.1 Run manifest	36
12.2 Suggested tabular event format	36
12.3 No hidden interpolation	37
13 Controls, Failure Modes and Falsification	38
13.1 The “alpha is only a grid size” objection	38
13.2 Finite precision is not the same as finite ontology	38
13.3 Numerical instability	38
13.4 Threshold dependence	39
13.5 Time-step dependence	39
13.6 Mesh orientation and graph dependence	39
13.7 A finite graph can still blow up numerically	39

14 Analysis Workflow	40
14.1 A complete reproducible study sequence	40
14.2 Minimal first experiment	41
15 Discussion and Intended Contribution	42
15.1 What the study hopes to achieve	42
15.2 Possible outcome I: smooth finite saturation	42
15.3 Possible outcome II: uncertainty replaces refinement	43
15.4 Possible outcome III: finite-state instability remains	43
15.5 Possible outcome IV: no distinctive FSM effect	43
15.6 What would count as a strong result	43
15.7 Relation to classical mathematics	44
15.8 Relation to Finite Symbolic Dynamics	44
15.9 The central hoped-for clarification	44
16 Recommended Deliverables	46
17 Conclusion	47
A Suggested Initial Parameter File	48
B Suggested Analysis Pseudocode	50
C Glossary	51

Chapter 1

Introduction and Use Case

1.1 Why Navier–Stokes is an unusually useful FSM test

Fluid motion offers a particularly clear use case for Finite Symbolic Mechanics because the classical theory openly crosses a representational boundary. A physical fluid is understood to possess microscopic structure, while the equations used at ordinary engineering scales replace that structure with smooth fields defined over a continuum. The replacement is extraordinarily successful. It allows velocity, density, pressure and viscosity to be organised into compact differential equations that predict and explain a vast range of observable behaviour.

The same move also imports the mathematical structure of the continuum. Once a field is taken to be defined at every point of $\mathbb{R}^3$, spatial derivatives are defined by limiting processes in which separations can be made arbitrarily small. Questions about smoothness, differentiability and singularity then become questions about the behaviour of the continuum representation under indefinite refinement.

FSM approaches the problem from the opposite direction. It begins with the finite act of distinction. If there is a smallest admissible distinguishable relation, represented here by the Alphonc limit α , then a physical or symbolic description cannot require

$$\Delta x \rightarrow 0 \tag{1.1}$$

as an executable operation. There may be increasingly fine finite distinctions, but the trajectory of refinement terminates at

$$\Delta x \geq \alpha. \tag{1.2}$$

This does not immediately make the Navier–Stokes equations wrong. It changes their status. They can be treated as a continuum compression of finite relations rather than as the irreducible form of those relations. That distinction creates a practical research question: if an incompressible flow is reconstructed as a finite conservation system whose refinement stops at α , what happens in regimes where the continuum formulation develops increasingly fine spatial structure?

The proposed study is therefore deliberately narrower than the classical Millennium Prize problem and, in another sense, more physically explicit. It does not seek a proof over an infinitely divisible domain. It seeks a reproducible finite experiment in which the boundary between resolvable structure and non-admissible requested refinement is made visible and measurable.

1.2 The immediate source of the question

A recent popular account of the Navier–Stokes problem makes the representational transition especially explicit. It notes that fluids are composed of atoms and molecules, but that fluid dynamics normally avoids tracking them individually and instead adopts a continuum assumption in which the fluid is treated as smooth. That smoothness is what makes the machinery of differential calculus directly available. The same account then identifies the unresolved mathematical problem as the question of whether sufficiently smooth three-dimensional solutions always remain smooth and globally defined, or whether a finite-time singularity can arise.

From an FSM perspective, those two statements should be placed beside one another. The continuum assumption is introduced as a useful modelling decision, and the famous unresolved problem is subsequently stated in terms of properties that belong to that continuum mathematical representation. The proposed research programme investigates whether this sequence is merely incidental or whether part of the difficulty can be localised to the assumptions that permit indefinite refinement. This motivating account is cited here as the immediate use-case source rather than as the mathematical authority for the Millennium problem [2].

1.3 The claim that is not being made

It is essential to separate three propositions.

First, classical Navier–Stokes theory is experimentally and computationally successful over enormous ranges of scale and application. Nothing in the present programme requires denying that fact.

Second, the Clay Mathematics Institute problem is a rigorous question posed within a continuum mathematical framework. A finite FSM reconstruction does not constitute a proof or disproof of that theorem unless it is independently shown to satisfy the precise classical hypotheses and conclusions. The present work therefore does not claim the Millennium Prize problem has been solved.

Third, it remains legitimate to ask whether the physical interpretation of the classical problem changes when arbitrary refinement is no longer admitted. This is the actual subject of the study.

The intended result is therefore a mapping between two questions:

$$\begin{array}{ll} \text{Classical: Can a smooth three-dimensional solution lose smoothness?} & \\ \text{FSM: Can a finite flow trajectory remain admissibly representable above } \alpha? & \end{array} \quad (1.3)$$

The research is valuable even if the two formulations ultimately remain mathematically independent. Showing precisely where they diverge would itself clarify the role of continuum assumptions in the classical problem.

Chapter 2

Historical and Mathematical Background

2.1 From continuum mechanics to Navier–Stokes

The modern equations emerged during the nineteenth century from attempts to combine Newtonian momentum balance with a constitutive account of internal fluid friction. Navier developed an early viscous-fluid formulation in the 1820s, while Stokes gave the theory a form recognisable to modern fluid mechanics in the 1840s. In later notation, the incompressible Newtonian equations are commonly written

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, \quad (2.1)$$

with the incompressibility constraint

$$\nabla \cdot \mathbf{u} = 0. \quad (2.2)$$

Here $\mathbf{u}(\mathbf{x}, t)$ is the velocity field, $p(\mathbf{x}, t)$ is pressure, ρ is density, ν is kinematic viscosity, and $\mathbf{f}$ is an applied body acceleration.

The nonlinear term

$$(\mathbf{u} \cdot \nabla) \mathbf{u} \quad (2.3)$$

expresses the transport of velocity by the velocity field itself. It is one of the central mathematical sources of the complexity of three-dimensional flow.

2.2 Weak solutions and the twentieth-century problem

In 1934 Jean Leray established the existence of global weak solutions for the three-dimensional incompressible equations under important classes of initial data. A weak solution relaxes the demand that all classical derivatives exist pointwise. This was a major advance, but it left open whether such solutions are always smooth and unique.

Subsequent work has established extensive regularity results, partial regularity results, criteria for smoothness and complete results in two dimensions. Nevertheless, the general three-dimensional existence and smoothness question remains open. The Clay Mathematics Institute

included the problem among the seven Millennium Prize Problems announced in 2000. As of August 2026, the Institute continues to list Navier–Stokes existence and smoothness as unresolved [1].

The difficulty therefore should not be summarised as “we cannot solve Navier–Stokes”. Many exact solutions are known in special cases, many numerical solutions are routinely obtained, and weak solutions are known to exist under broad conditions. The unresolved question is more precise: whether smooth, divergence-free initial data necessarily produce a globally smooth solution in the required three-dimensional setting, or whether finite-time breakdown can occur.

2.3 The role of the continuum assumption

The continuum model replaces microscopic discreteness with smooth macroscopic fields. In the usual interpretation this is justified when the scale of interest is much larger than molecular mean free paths and when local averages behave reproducibly. The field model is consequently an effective description rather than a literal molecular inventory.

The mathematical continuum, however, has stronger properties than a finite physical averaging procedure. If $u(x)$ is differentiable, then

$$\frac{\partial u}{\partial x} = \lim_{h \rightarrow 0} \frac{u(x+h) - u(x)}{h}. \quad (2.4)$$

No finite measurement performs the limit. Instead, finite observations are interpreted through a mathematical structure in which the limit is well-defined.

FSM does not need to deny the utility of that structure. It asks whether the explanatory direction should instead be

$$\text{finite registrations} \longrightarrow \text{finite relations} \longrightarrow \text{continuum compression}, \quad (2.5)$$

rather than

$$\text{continuum field} \longrightarrow \text{finite discretisation} \longrightarrow \text{approximate computation}. \quad (2.6)$$

That reversal is the central conceptual move of the present research plan.

Chapter 3

Finite Symbolic Mechanics Framework

3.1 Finite symbolic representation

FSM begins from the proposition that a usable scientific representation is finite. A measured quantity is not treated as an infinitely precise real number that has merely been imperfectly observed. It is a finite symbolic registration with a finite distinction, a context, an uncertainty and a history of production.

For the present study, a registered scalar may be written schematically as

$$X_i^n = (x_i^n, \delta x_i^n, H_i^n), \quad (3.1)$$

where x_i^n is the reported value, δx_i^n represents the admissible uncertainty or resolution associated with it, and H_i^n is a provenance record. The provenance need not be metaphysical. In a numerical experiment it may contain the mesh identifier, update rule, parent state, interpolation history, boundary condition, solver version and parameter set from which the symbol was produced.

3.2 The Alphonic limit

The Alphonic limit α is the lower bound on admissible spatial distinction in the proposed FSM representation. If two proposed spatial registrations differ by less than α , the model does not treat them as two independently available spatial symbols.

Thus

$$d_{ij} \geq \alpha \quad (3.2)$$

for independently represented neighbouring locations or cell extents in the minimal model.

This research plan does not assume that the numerical value of a physically universal α has already been established. The first computational studies should treat the dimensionless ratio

$$\alpha^* = \frac{\alpha}{L} \quad (3.3)$$

as an experimental parameter, where L is the characteristic system length. This is important because a macroscopic fluid simulation cannot computationally resolve a picometre-scale physical cutoff directly. The first task is to study the consequences of a finite representational floor as such. Calibration to a physical Alphonic scale is a later and separate empirical question.

3.3 Finite time

A spatial Alphonic limit alone does not justify an infinitely divisible temporal update. The implementation should therefore use a finite registered time step Δt . If the wider FSM programme assigns a temporal Alphonic limit τ_α , the additional admissibility condition is

$$\Delta t \geq \tau_\alpha. \quad (3.4)$$

For the initial research programme, Δt should be determined by numerical stability, temporal-resolution requirements and the finite time-registration policy. The spatial and temporal limits should not be conflated unless an independent physical relation between them has been established.

3.4 Uncertainty and provenance

Two simulations with the same displayed state can have different epistemic status if they arose through different refinement histories or numerical operations. FSM therefore requires the computational record to preserve more than the visible velocity field.

Each cell should minimally carry

$$S_i^n = \{m_i^n, \mathbf{P}_i^n, p_i^n, \delta_i^n, H_i^n\}, \quad (3.5)$$

where m_i is represented mass, $\mathbf{P}_i$ is represented momentum, p_i is a pressure-like constraint variable, δ_i is an uncertainty record, and H_i is provenance.

The velocity is a derived finite relation

$$\mathbf{u}_i^n = \frac{\mathbf{P}_i^n}{m_i^n}, \quad (3.6)$$

provided m_i^n is admissibly non-zero.

3.5 Admissibility

An FSM state is not accepted merely because a floating-point program returned a number. It must pass declared admissibility conditions. For the present programme these should include:

1. every represented cell or independent spatial separation satisfies the Alphonic lower bound;
2. mass and momentum transfers are finite and locally accounted for;

3. the incompressibility constraint is satisfied within a declared finite tolerance;
4. uncertainty and provenance are retained through updates and refinement;
5. no refinement operation creates a child state whose characteristic extent is smaller than α ;
6. any requested but prohibited refinement is recorded explicitly rather than silently ignored.

The last requirement is the distinctive experimental feature. A failure of resolution becomes an observable output of the model.

Chapter 4

A Finite Graph Formulation of Fluid Motion

4.1 Why begin with a graph rather than a discretised derivative

If the purpose of the study is foundational, it would be circular to define the FSM model by first writing continuum derivatives and then merely replacing them by finite differences. That procedure is entirely legitimate for computational fluid dynamics, but it would leave the continuum as the prior object.

A cleaner FSM construction begins with a finite adjacency graph

$$G = (V, E), \quad (4.1)$$

where each vertex $i \in V$ represents a finite cell C_i and each edge $e = (i, j) \in E$ represents a permitted interaction across a shared face or finite neighbourhood relation.

The geometry associated with each edge contains only finite quantities: face area A_e , centre-to-centre separation d_e , orientation $\mathbf{n}_e$ and an admissibility condition

$$d_e \geq \alpha. \quad (4.2)$$

Differential operators can later be recovered as continuum interpretations of these finite relations, but the computational system itself need not invoke a limit to zero.

4.2 Incidence operators

Let B be the oriented vertex-edge incidence matrix of the graph. An edge flux vector $\mathbf{q}$ is mapped to a net cell balance by

$$D_G \mathbf{q} = B \mathbf{q}, \quad (4.3)$$

where D_G is a finite divergence-like operator.

A cell scalar p can be mapped to oriented edge differences by

$$G_G p = -B^\top p, \quad (4.4)$$

with geometric weights introduced separately. A graph Laplacian is then

$$L_G = D_G W G_G, \quad (4.5)$$

where W is a diagonal matrix of finite geometric weights.

These are finite matrices on a finite graph. Their use does not require the assertion that a derivative exists at a zero-extent point.

4.3 Mass balance

For a general finite cell, mass evolves by face transfers

$$m_i^{n+1} = m_i^n - \Delta t \sum_{e \in \partial C_i} \sigma_{ie} F_e^{(m)}, \quad (4.6)$$

where $F_e^{(m)}$ is the finite mass-transfer rate through edge or face e and $\sigma_{ie} \in \{-1, +1\}$ records orientation.

For the constant-density incompressible experiments proposed later, the stronger working constraint is that net volume flux into each cell vanishes to a declared tolerance:

$$D_G \mathbf{q}^{n+1} \approx \mathbf{0}. \quad (4.7)$$

The tolerance must be reported. It must not be hidden behind the statement that the divergence is “zero”.

4.4 Momentum balance

The finite momentum update can be written schematically as

$$\mathbf{P}^{n+1} = \mathbf{P}^n + \Delta t [-\mathcal{A}_G(\mathbf{U}^n) - \mathcal{G}_G(\mathbf{p}^n) + \nu \mathcal{L}_G(\mathbf{U}^n) + \mathbf{F}^n], \quad (4.8)$$

where $\mathcal{A}_G$ is a nonlinear finite advection/transport operator, $\mathcal{G}_G$ is a finite pressure-difference operator, $\mathcal{L}_G$ is a finite viscous exchange operator, and $\mathbf{F}^n$ is a finite external contribution.

The crucial point is interpretive. Equation (4.8) is not required to be described as a poor approximation to an unseen infinitely resolved field. It is the directly executable finite model. Its relation to Equation (2.1) becomes an empirical and mathematical comparison.

4.5 Finite advection

For each oriented face $e = (i, j)$, define a finite face velocity $\mathbf{u}_e$ from admissible neighbouring states. A conservative mass or momentum flux can be constructed with an upwind, central, flux-limited or Riemann-type rule. The exact rule should be varied in the robustness study because a boundary event that occurs only under one transport closure may be a property of that closure rather than of the Alphoncic constraint.

The nonlinear character of the flow remains. For example, a face momentum transfer may contain products of finite velocity and finite momentum flux. FSM therefore removes neither nonlinearity nor computational difficulty. It removes only the requirement that the nonlinear relation be grounded in an infinitely refinable derivative.

4.6 Finite viscosity

Viscous interaction can be represented by a weighted neighbour exchange

$$\mathcal{V}_i = \nu \sum_{j \in N(i)} w_{ij} (\mathbf{u}_j - \mathbf{u}_i), \quad (4.9)$$

where the weights w_{ij} are finite functions of face area, cell volume and finite separation. On a regular grid this becomes algebraically equivalent to familiar finite-volume diffusion operators. The difference remains foundational: Equation (4.9) is accepted as a finite neighbour relation rather than justified by taking a second derivative at a mathematical point.

4.7 Pressure as a finite constraint relation

A conventional projection method often derives a Poisson equation for pressure from the continuum equations. The FSM implementation can avoid making the continuum Poisson equation foundational by defining pressure correction as the finite algebraic operation needed to enforce Equation (4.7).

If an unconstrained provisional face flux is $\mathbf{q}^*$, seek a finite scalar correction ϕ such that

$$\mathbf{q}^{n+1} = \mathbf{q}^* - WG_G \phi \quad (4.10)$$

and

$$D_G \mathbf{q}^{n+1} = \mathbf{0} \quad (4.11)$$

within the declared tolerance. Therefore

$$D_G W G_G \phi = D_G \mathbf{q}^*. \quad (4.12)$$

Equation (4.12) is a finite linear system on the graph. It resembles a discretised pressure-Poisson equation because the same conservation structure is present, but no limiting derivative is necessary to define the operation.

4.8 Finite gradient and an Alphonic bound

A conventional gradient between neighbouring represented states can be replaced by a finite relation

$$D_\alpha u_{ij} = \frac{u_j - u_i}{d_{ij}}, \quad d_{ij} \geq \alpha. \quad (4.13)$$

This quantity can become large, but it cannot become infinite merely because d_{ij} tends to zero: the denominator is not permitted to do so. Consequently a continuum statement such as

$$|\nabla \mathbf{u}| \rightarrow \infty \quad (4.14)$$

requires an FSM reinterpretation. The relevant operational question is whether increasing state variation would require a new distinguishable region with extent below α in order to maintain the declared representation standard.

Chapter 5

The Alphonic Boundary Experiment

5.1 Defining a boundary event

The phrase “the dynamics request finer resolution” must be operationalised. Otherwise it risks becoming a metaphor. The proposed implementation therefore defines one or more finite refinement indicators.

For a velocity component, a simple local variation indicator is

$$R_i^{(u)} = \max_{j \in N(i)} \frac{\|\mathbf{u}_j - \mathbf{u}_i\|}{u_{\text{scale}} + \delta u_i + \delta u_j}, \quad (5.1)$$

where u_{scale} prevents division by an arbitrarily small normalising value and should be fixed by the experimental protocol.

A pressure indicator may be defined similarly,

$$R_i^{(p)} = \max_{j \in N(i)} \frac{|p_j - p_i|}{p_{\text{scale}} + \delta p_i + \delta p_j}. \quad (5.2)$$

A finite conservation indicator can be based on residual flux,

$$R_i^{(c)} = \frac{|(D_G \mathbf{q})_i|}{q_{\text{scale}}}. \quad (5.3)$$

The local refinement demand is then

$$R_i = \max \left(R_i^{(u)}, R_i^{(p)}, R_i^{(c)} \right). \quad (5.4)$$

If $R_i > R_{\text{crit}}$ and the cell can be refined without violating the Alphonic limit, refinement is permitted. If $R_i > R_{\text{crit}}$ and refinement would require a child extent below α , the code records an Alphonic boundary event (ABE).

5.2 An Alphonic boundary event is data

Define

$$A_i^n = \begin{cases} 1, & R_i^n > R_{\text{crit}} \text{ and no admissible refinement exists,} \\ 0, & \text{otherwise.} \end{cases} \quad (5.5)$$

The total event count at time n is

$$N_{\text{ABE}}(n) = \sum_i A_i^n, \quad (5.6)$$

and the occupied Alphonic-boundary fraction is

$$f_{\text{ABE}}(n) = \frac{\sum_i V_i A_i^n}{\sum_i V_i}. \quad (5.7)$$

The model should retain the spatial location, neighbouring states, uncertainty, refinement history and cause of every ABE. The event must not automatically be called a “physical singularity”. It is first and foremost a statement about the relation between the evolving finite model and its own declared representational floor.

5.3 Three possible responses at the boundary

The first implementation should not hard-code one philosophical answer to an ABE. It should support at least three experimental policies.

Policy A: strict refusal. The model refuses sub- α refinement and continues with the existing finite interactions. The unresolved local variation is retained explicitly.

Policy B: uncertainty expansion. The model refuses refinement but increases the uncertainty attached to the affected state. The increase is recorded as representational loss rather than smoothed away.

Policy C: conservative redistribution. The model applies only the already-defined finite viscous and conservative transfer rules, allowing neighbouring states to redistribute momentum without creating additional spatial symbols.

Comparing these policies is important. If all produce the same large-scale trajectory and similar ABE structure, the boundary effect is robust. If results depend strongly on the policy, then the policy forms part of the model and must be reported as such.

Chapter 6

Research Questions, Hypotheses and Outcomes

6.1 Primary research question

The primary research question is:

When incompressible three-dimensional flow is represented as a finite conservative interaction system with a minimum admissible spatial distinction α , how does the system behave when local variation reaches the representational floor, and how does that behaviour compare with the corresponding continuum-based numerical model under progressive refinement?

6.2 Secondary questions

The investigation should answer the following subsidiary questions.

1. Do ABE events appear reproducibly across numerical transport rules, time-step choices and mesh topologies?
2. Do ABE events cluster around conventional high-vorticity or high-strain regions?
3. Does the frequency or spatial organisation of ABE events scale systematically with Reynolds number and with α/L ?
4. Does the large-scale trajectory become insensitive to further classical refinement before the Alphonic floor is reached?
5. When the Alphonic floor is reached, does the finite system remain bounded under all tested initial conditions?
6. If finite state variables become numerically unbounded despite a non-zero spatial denominator, what mechanism produces that growth?
7. Does uncertainty expansion absorb unresolved structure, or does it spread until the model loses useful predictive distinction?

8. Can nonlinear dynamical diagnostics identify a transition associated with approaching the Alphonic boundary?

6.3 Working hypotheses

The research should begin with hypotheses that can fail.

H1: finite-denominator hypothesis. A finite lower spatial bound prevents divergence that arises solely from the denominator of a spatial difference tending to zero.

H2: boundary-substitution hypothesis. Some continuum singularity-like behaviour will appear in the finite model as increased ABE density, increased finite neighbour contrast, increased uncertainty or a change in trajectory structure rather than as an infinite local gradient.

H3: scale-robustness hypothesis. Above the Alphonic boundary, large-scale observables will converge across successively refined finite representations in the same practical sense expected of a competent conservative numerical method.

H4: model-dependence test. If ABE behaviour varies strongly with flux closure, mesh topology or threshold definition, the phenomenon should be attributed primarily to the chosen finite representation rather than to a general FSM constraint.

H5: no-free-regularity hypothesis. Imposing $\alpha > 0$ does not by itself guarantee every state variable remains bounded. Any claim of regularisation must therefore be demonstrated by the update dynamics, not inferred merely from finite spatial resolution.

H5 is particularly important. A finite graph can still support rapidly growing or even numerically divergent state values. The research must test this directly rather than assuming that eliminating $h \rightarrow 0$ automatically proves boundedness.

Chapter 7

Experimental Programme

7.1 Stage 0: implementation verification

Before any claim about FSM is considered, the finite solver must pass ordinary computational checks. These include mass conservation, momentum accounting, symmetry preservation where expected, reproduction of simple known flows, time-step convergence over a range where the Alphonic cutoff is inactive, and independence from irrelevant ordering of graph nodes.

A failed validation is a solver failure, not evidence for FSM.

7.2 Stage 1: two-dimensional validation problems

Two-dimensional problems are not direct tests of the three-dimensional Millennium question, but they provide valuable verification because two-dimensional incompressible Navier–Stokes behaviour is mathematically better controlled.

Recommended initial cases are:

- a. periodic decaying vortices with smooth divergence-free initial velocity;
- b. a lid-driven cavity as a boundary-condition test;
- c. a shear-layer roll-up problem as a controlled test of increasing local structure.

The purpose is to verify conservation, graph projection, refinement, ABE logging and uncertainty handling before entering the three-dimensional study.

7.3 Stage 2: three-dimensional Taylor–Green vortex

The Taylor–Green vortex is recommended as the primary reproducible three-dimensional test because it begins from a smooth, divergence-free field and develops progressively smaller structures as nonlinear interactions evolve.

On a periodic cube, a standard form is

$$u_x(x, y, z, 0) = U_0 \sin x \cos y \cos z, \quad (7.1)$$

$$u_y(x, y, z, 0) = -U_0 \cos x \sin y \cos z, \quad (7.2)$$

$$u_z(x, y, z, 0) = 0. \quad (7.3)$$

The precise domain scaling can be chosen so that $L = 2\pi$ or nondimensionalised to $L = 1$. What matters is that all parameters and transformations are recorded.

A proposed first Reynolds-number sweep is

$$Re \in \{100, 400, 800, 1600\}, \quad (7.4)$$

with the explicit understanding that these are research-plan values and may be changed after validation. The Reynolds number is

$$Re = \frac{U_0 L}{\nu}. \quad (7.5)$$

7.4 Stage 3: Alphonic refinement series

For each initial condition and Reynolds number, run a family of finite representations with

$$\alpha^* = \frac{\alpha}{L} \in \left\{ \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \frac{1}{128} \right\}, \quad (7.6)$$

where computational resources permit.

The study should distinguish the global Alphonic floor from the current local cell size. A cell of size 8α may refine to 4α , then 2α , then α , but no rule may create a child cell smaller than α .

For each α^* , run an otherwise matched classical adaptive simulation that is allowed at least one or two further refinement levels. This provides a direct comparison between

$$\text{refinement permitted} \quad \text{and} \quad \text{refinement prohibited by } \alpha. \quad (7.7)$$

7.5 Stage 4: smooth high-shear initial-condition family

A single canonical vortex is not enough. Construct a family of smooth, divergence-free initial conditions with controlled increasing shear or spectral concentration while keeping total initial quadratic velocity norm comparable. The goal is to ask whether the approach to the Alphonic boundary depends systematically on the distribution of finite structure rather than on a single specially chosen flow.

All initial conditions must be generated by a reproducible seed or closed analytic expression. Random fields should be accompanied by their exact random seed and generation parameters.

7.6 Stage 5: mesh and topology robustness

Repeat selected cases on at least two finite geometries, for example:

- a. a regular Cartesian finite-volume graph;
- b. a mildly irregular or adaptively refined graph preserving comparable cell volumes.

If the proposed Alphonic effect is genuine at the level of the finite framework, its gross signatures should not depend on one particular lattice orientation.

7.7 Stage 6: uncertainty study

After the deterministic solver is stable, introduce finite uncertainty into initial velocity and model parameters. A computationally transparent first method is an ensemble interval procedure rather than a differential error-propagation formula.

For each nominal initial state S^0 , construct perturbed states

$$S^{0,(k)} \in \mathcal{U}(S^0, \delta S^0) \tag{7.8}$$

within declared finite bounds. Evolve the ensemble and record the spread of observables, ABE locations and reconstructed trajectories. This prevents a hidden assumption that exact floating-point initial data represent exact physical knowledge.

Chapter 8

Computational Architecture

8.1 Recommended software structure

A minimal Python research code should be modular enough that the FSM assumptions can be altered without rewriting the flow solver. The following logical components are recommended:

1. **Geometry:** cells, faces, graph adjacency, finite areas and distances;
2. **State:** mass, momentum, pressure relation, uncertainty and provenance;
3. **Flux:** advection and viscous exchange rules;
4. **Constraint:** finite graph pressure correction and incompressibility residual;
5. **Admissibility:** Aliphonic limit, refinement criteria and boundary-event logger;
6. **Recorder:** time-series, fields, event logs and parameter metadata;
7. **Analysis:** conventional diagnostics and nonlinear dynamical methods;
8. **Visualisation:** snapshots, trajectories, spectra, recurrence plots and event maps.

The research code should store configuration files alongside each run. A result without its α , time-step policy, flux rule, graph version and thresholds should be treated as incomplete.

8.2 Core data structures: Python-oriented pseudocode

```
@dataclass
class CellState:
    cell_id: int
    volume: float
    centre: array
    extent: float
    mass: float
    momentum: array
    pressure: float
    uncertainty: dict
```

```

    provenance: dict

    @property
    def velocity(self):
        assert self.mass > 0
        return self.momentum / self.mass

@dataclass
class FSMConfig:
    alpha: float
    dt: float
    viscosity: float
    rho: float
    refinement_threshold: float
    divergence_tolerance: float
    boundary_policy: str
    flux_scheme: str
    seed: int

@dataclass
class BoundaryEvent:
    step: int
    time: float
    cell_id: int
    extent: float
    indicator: float
    cause: str
    local_state: dict
    uncertainty: dict
    provenance: dict

```

Listing 8.1: Finite cell and run-state records.

8.3 Building an admissible graph

```

def build_graph(domain, base_resolution, config):
    cells, faces = create_finite_cells(domain, base_resolution)

    for cell in cells:
        if cell.extent < config.alpha:
            raise ValueError("Initial graph violates Alphonc limit")

    for face in faces:
        if centre_distance(face.left, face.right) < config.alpha:
            raise ValueError("Independent nodes are closer than alpha")

    B = incidence_matrix(cells, faces)
    W = geometric_weights(cells, faces)

    return Graph(cells=cells, faces=faces, B=B, W=W)

```

Listing 8.2: Graph construction with an Alphonc constraint.

8.4 Finite transport and viscous exchange

```
def compute_face_transfers(graph, state, config):
    transfers = []

    for face in graph.faces:
        left = state[face.left]
        right = state[face.right]

        u_face = admissible_face_velocity(
            left.velocity,
            right.velocity,
            scheme=config.flux_scheme
        )

        mass_flux = finite_mass_flux(face, u_face, config.rho)

        advective_momentum = finite_momentum_transport(
            face, left, right, u_face, mass_flux
        )

        viscous_momentum = finite_viscous_exchange(
            face,
            left.velocity,
            right.velocity,
            config.viscosity
        )

        transfers.append(
            FaceTransfer(
                mass=mass_flux,
                momentum=advective_momentum + viscous_momentum
            )
        )

    return transfers
```

Listing 8.3: Schematic finite face transfers.

8.5 Finite incompressibility projection

```
def enforce_incompressibility(graph, provisional_flux, config):
    # Finite residual at each cell
    residual = graph.B @ provisional_flux

    # Graph Laplacian: a finite algebraic operator
    Lg = graph.B @ graph.W @ (-graph.B.T)

    # Gauge condition or reference pressure is required because
    # pressure is defined only up to an additive constant.
    Lg_fixed, rhs_fixed = apply_pressure_gauge(Lg, residual)

    phi = sparse_solve(Lg_fixed, rhs_fixed)
```

```

corrected_flux = (
    provisional_flux
    - graph.W @ (-graph.B.T) @ phi
)

final_residual = graph.B @ corrected_flux

if max_abs(final_residual) > config.divergence_tolerance:
    log_constraint_warning(final_residual)

return corrected_flux, phi, final_residual

```

Listing 8.4: Graph-based pressure correction.

8.6 Refinement and Alphonic refusal

```

def evaluate_and_refine(graph, state, step, time, config):
    events = []
    cells_to_refine = []

    for cell in graph.cells:
        R_u = velocity_variation_indicator(cell, graph, state)
        R_p = pressure_variation_indicator(cell, graph, state)
        R_c = conservation_indicator(cell, graph, state)
        R = max(R_u, R_p, R_c)

        if R <= config.refinement_threshold:
            continue

        proposed_extent = child_extent(cell)

        if proposed_extent >= config.alpha:
            cells_to_refine.append(cell.cell_id)
        else:
            events.append(
                BoundaryEvent(
                    step=step,
                    time=time,
                    cell_id=cell.cell_id,
                    extent=cell.extent,
                    indicator=R,
                    cause=dominant_indicator(R_u, R_p, R_c),
                    local_state=snapshot_neighbourhood(cell, graph, state),
                    uncertainty=state[cell.cell_id].uncertainty,
                    provenance=state[cell.cell_id].provenance
                )
            )

    graph, state = refine_admissible_cells(
        graph, state, cells_to_refine, alpha=config.alpha
    )

    state = apply_boundary_policy(state, events, config.boundary_policy)

    return graph, state, events

```

8.7 Full simulation loop

```
def run_fsm_flow(domain, initial_state, config, t_end):
    graph = build_graph(domain, base_resolution="coarse", config=config)
    state = initialise_state(graph, initial_state, config)
    recorder = Recorder(config=config)

    t = 0.0
    step = 0

    while t < t_end:
        dt = choose_finite_dt(graph, state, config)

        if has_temporal_alphonic_limit(config):
            assert dt >= config.tau_alpha

        transfers = compute_face_transfers(graph, state, config)

        provisional_state, provisional_flux = conservative_update(
            graph, state, transfers, dt, config
        )

        corrected_flux, phi, div_residual = enforce_incompressibility(
            graph, provisional_flux, config
        )

        state = apply_flux_correction(
            graph, provisional_state, corrected_flux, phi, dt
        )

        state = propagate_finite_uncertainty(
            graph, state, transfers, dt, config
        )

        graph, state, events = evaluate_and_refine(
            graph, state, step, t, config
        )

        diagnostics = calculate_diagnostics(
            graph, state, events, div_residual, config
        )

        recorder.store(
            step=step,
            time=t,
            graph=graph,
            state=state,
            events=events,
            diagnostics=diagnostics
        )
```

```

    assert no_cell_below_alpha(graph, config.alpha)
    assert finite_accounting_is_valid(graph, state, config)

    t += dt
    step += 1

return recorder.finalise()

```

Listing 8.6: High-level FSM flow simulation.

8.8 Matched classical-refinement control

The same codebase should support a control mode in which the minimum cell size is allowed to fall below the nominated FSM α . This does not make the control mathematically continuous; all computation remains finite. Its purpose is practical: it shows what additional structure the conventional numerical method attempts to represent when the FSM run refuses the next level of refinement.

```

for case in experiment_cases:
    fsm_result = run_case(
        case,
        alpha=case.alpha,
        stop_refinement_at_alpha=True
    )

    control_result = run_case(
        case,
        alpha=case.alpha,
        stop_refinement_at_alpha=False,
        extra_refinement_levels=2
    )

    compare_large_scale_observables(fsm_result, control_result)
    compare_high_variation_regions(fsm_result, control_result)
    compare_trajectory_diagnostics(fsm_result, control_result)

```

Listing 8.7: Matched refinement comparison.

Chapter 9

Measurements and Diagnostics

9.1 Core quantities to record

Every run should retain both full fields at selected checkpoints and compact time-series diagnostics at every time step. Recommended primary diagnostics are shown in Table 9.1.

Table 9.1: Primary diagnostics for the FSM Navier–Stokes study.

Diagnostic	Purpose
Maximum finite neighbour velocity contrast	Detects locally increasing state separation without invoking $h \rightarrow 0$.
Maximum $\ D_\alpha \mathbf{u}\ $	Finite analogue of a strong local gradient.
Divergence residual	Checks the finite incompressibility constraint.
Total represented mass	Conservation check.
Total represented momentum	Accounting and boundary-condition check.
Quadratic velocity norm $\sum_i V_i \ \mathbf{u}_i\ ^2$	Classical comparison diagnostic without requiring a point-wise continuum field.
Finite circulation / vorticity proxy	Locates rotating structures using finite loops or neighbour differences.
ABE count N_{ABE}	Number of unresolved refinement demands.
ABE volume fraction f_{ABE}	Spatial prevalence of boundary saturation.
ABE persistence time	Distinguishes transient refinement pressure from persistent representational saturation.
Uncertainty volume or ensemble spread	Tracks loss of finite distinction under unresolved dynamics.
Refinement depth	Records the trajectory by which each region reached α .
Conservation residuals	Distinguishes physical-model effects from numerical failure.

9.2 Finite circulation rather than point vorticity

Vorticity is conventionally defined by

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}. \quad (9.1)$$

For an FSM-first implementation, a more direct finite quantity is circulation around a finite cell loop Γ_i :

$$\Gamma_i^{(u)} = \sum_{e \in \partial C_i} \mathbf{u}_e \cdot \Delta \mathbf{l}_e. \quad (9.2)$$

A volume- or area-normalised circulation can then be used as a finite rotational diagnostic. Conventional vorticity may still be calculated for comparison, but it should be labelled as a derived continuum-style diagnostic rather than the foundational state variable.

9.3 A boundedness audit

At each time step the code should explicitly audit

$$\max_i \|\mathbf{u}_i\|, \quad \max_i |p_i|, \quad \max_{e=(i,j)} \frac{\|\mathbf{u}_j - \mathbf{u}_i\|}{d_{ij}}. \quad (9.3)$$

If any quantity grows rapidly, the run should record whether the cause is physical transport, pressure correction, numerical instability, boundary policy, loss of mass positivity or accumulation of uncertainty. This is necessary because a finite spatial graph does not logically guarantee bounded state amplitudes.

Chapter 10

Visualisation Plan

10.1 Visualisation as part of the method

The visual programme should not be decorative. It should make the difference between classical refinement and FSM admissibility inspectable. Wherever possible, the same camera, slice positions, normalisation and time points should be used for matched runs.

10.2 Figure set A: conceptual framing

The opening conceptual figure should contrast the two explanatory trajectories.

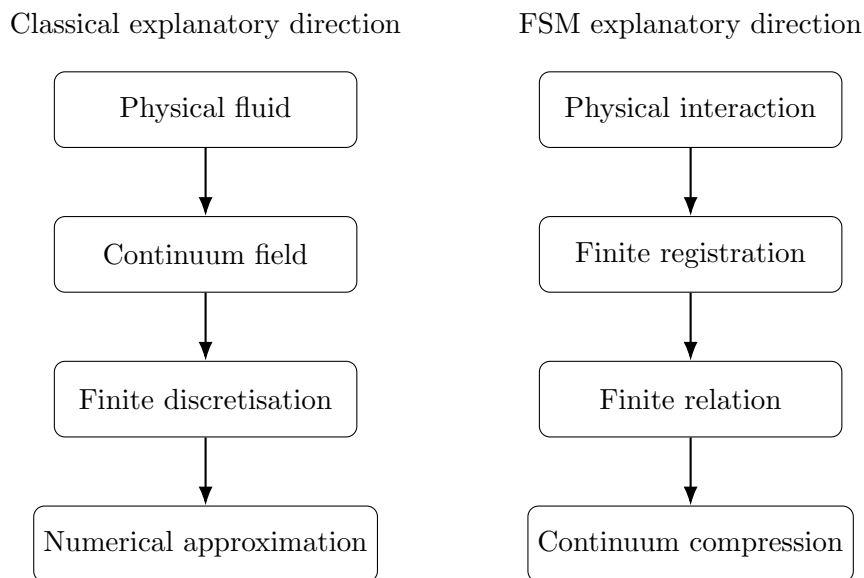


Figure 10.1: The proposed research does not discard the continuum description; it reverses the proposed foundational direction.

A second conceptual figure should show the refinement ladder

$$8\alpha \rightarrow 4\alpha \rightarrow 2\alpha \rightarrow \alpha \rightarrow \boxed{\text{no further spatial distinction}}. \quad (10.1)$$

10.3 Figure set B: flow fields

For every principal three-dimensional case generate:

- a. orthogonal velocity-magnitude slices;
- b. finite-circulation or conventional-vorticity slices;
- c. pressure-relation slices;
- d. refinement-level maps;
- e. ABE location maps;
- f. uncertainty maps.

A useful composite visual is a slice coloured by refinement level with ABE cells outlined. This directly shows whether unresolved finite structure clusters in dynamically active regions.

10.4 Figure set C: time-series plots

The following plots are particularly important:

- a. $N_{\text{ABE}}(t)$ and $f_{\text{ABE}}(t)$;
- b. maximum finite neighbour contrast versus time;
- c. maximum finite gradient proxy versus time;
- d. divergence residual versus time;
- e. uncertainty spread versus time;
- f. quadratic velocity norm versus time;
- g. matched FSM/control differences in large-scale observables versus time.

The first ABE time should be marked on all relevant time series. This permits direct inspection of whether trajectory changes precede, coincide with or follow representational saturation.

10.5 Figure set D: Alphonic scaling plots

For each Reynolds number, plot summary quantities against α/L on logarithmic axes where appropriate. Candidate summaries include first ABE time, peak ABE fraction, peak finite neighbour contrast, final uncertainty and mismatch from the deeper-refined control.

The purpose is not merely to show that finer grids give different answers. It is to identify whether a reproducible scaling relation exists between the finite representational floor and the onset of unresolved local structure.

10.6 Figure set E: trajectory visualisations

For selected scalar observables $s(t)$, construct delay-coordinate plots

$$\mathbf{y}(t) = [s(t), s(t - \tau), s(t - 2\tau)]. \quad (10.2)$$

Colour or annotate trajectory segments according to ABE density. This creates a direct bridge between FSM, Finite Symbolic Dynamics and nonlinear dynamical analysis.

Chapter 11

Nonlinear Dynamical Analysis

11.1 Why nonlinear analysis is useful here

A field comparison alone may show where the finite and classical-control calculations differ but not whether the global organisation of the system has changed. Nonlinear dynamical methods can examine whether approaching the Alphonic floor corresponds to a change in trajectory geometry, recurrence, predictability or effective dimensionality.

These methods should be applied to finite recorded observables. They do not require an ontological commitment to an infinite-dimensional continuum state space.

11.2 Delay-coordinate reconstruction

For a scalar observable s_n sampled at finite times, construct

$$\mathbf{y}_n = (s_n, s_{n-k}, s_{n-2k}, \dots, s_{n-(m-1)k}), \quad (11.1)$$

where k is an integer delay and m is an embedding dimension.

Candidate observables include:

- a. maximum finite neighbour contrast;
- b. total finite circulation norm;
- c. $N_{\text{ABE}}(t)$;
- d. uncertainty spread;
- e. a low-order spatial moment of the velocity field;
- f. flux through a fixed finite surface.

The delay should be selected by a finite data criterion such as first minimum of mutual information or a decorrelation time, and m can be explored with false-nearest-neighbour analysis. Because the data are finite, all embedding choices and thresholds must be reported.

11.3 Recurrence plots and recurrence quantification

For reconstructed states $\mathbf{y}_i$, define a finite recurrence matrix

$$R_{ij} = \Theta(\varepsilon - \|\mathbf{y}_i - \mathbf{y}_j\|), \quad (11.2)$$

where ε is a declared finite recurrence threshold.

Recurrence plots can reveal laminar intervals, intermittency, trajectory transitions and changes in predictability. Recurrence quantification measures such as recurrence rate, determinism, laminarity and diagonal-line statistics should be compared before and after the first ABE and between FSM and deeper-refined control runs.

This method is especially suitable for FSM because recurrence is itself a finite relational test rather than a demand for analytic differentiability.

11.4 Finite-time Lyapunov analysis

A finite-time Lyapunov exponent or related perturbation-growth measure can be estimated from pairs of nearby finite states or matched perturbed simulations. The objective is not to claim an exact asymptotic Lyapunov spectrum from finite data. It is to quantify whether neighbouring admissible trajectories separate more rapidly as the Alphoncic boundary is approached.

A direct ensemble measure is

$$\lambda_T = \frac{1}{T} \log \frac{d(T)}{d(0)}, \quad (11.3)$$

where $d(t)$ is a declared finite state-space distance between two initially nearby simulations. Use several initial perturbations and report the distribution rather than a single number.

11.5 Correlation dimension and scaling cautions

The Grassberger–Procaccia correlation-sum approach may be applied exploratorily to reconstructed trajectories. However, it must be handled cautiously. Apparent scaling plateaux can be created by short data, noise and non-stationarity. The purpose is therefore comparative: does the estimated effective dimension change systematically across α/L , Reynolds number and ABE onset?

The result should not be interpreted as proof that the full fluid “has” a particular fractal dimension.

11.6 Permutation entropy and symbolic complexity

Permutation entropy is particularly attractive because it converts a finite time series into ordinal patterns. For embedding order m and delay k , local sequences are mapped to one of finitely many rank-order symbols. The entropy of the resulting symbolic distribution can track changes in temporal complexity.

This provides a natural connection to FSD. One can ask whether the symbolic trajectory becomes more complex as local structure approaches α , and whether a boundary event is followed by saturation, redistribution or loss of distinguishability.

11.7 Recurrence networks

The recurrence matrix can also be interpreted as the adjacency matrix of a finite network whose nodes are reconstructed states. Network measures such as degree distribution, transitivity, clustering and path length can describe changes in the geometry of the sampled trajectory.

This may be particularly useful if the wider FSM programme treats trajectories as finite graph structures rather than as curves embedded in an assumed smooth manifold.

11.8 Poincare sections and return maps

For runs exhibiting quasiperiodic or recurrent large-scale behaviour, define a finite crossing condition on a chosen observable and record successive return values. A return map can reveal periodic windows, bifurcations and intermittent transitions. The crossing rule must include a finite tolerance so that it remains an executable symbolic operation.

11.9 Transfer entropy as an optional spatial diagnostic

For selected regions, transfer entropy can be used to estimate directed predictive dependence between finite cell or regional time series. This may help distinguish whether ABE clusters are merely local high-gradient markers or whether they sit in regions that strongly steer subsequent large-scale flow organisation.

This analysis is optional because it is data-intensive and sensitive to estimator choices. It should follow, not precede, the simpler conservation and recurrence analyses.

11.10 Linear comparison methods

Proper orthogonal decomposition and dynamic mode decomposition may still be useful as comparison tools even though they are not themselves nonlinear dynamical analyses in the same sense. They can provide low-order representations of the dominant spatial structures and help determine whether the FSM boundary changes the large-scale mode organisation. Any such analysis should be clearly labelled as a descriptive reduction, not as the nonlinear core of the study.

Chapter 12

Data, Provenance and Reproducibility

12.1 Run manifest

Every simulation should generate a machine-readable manifest containing at least:

- code version and repository commit;
- date and platform;
- domain geometry and boundary conditions;
- initial-condition identifier and random seed if applicable;
- α , Δt policy, viscosity, density and Reynolds number;
- flux scheme and pressure-constraint solver;
- refinement and coarsening rules;
- ABE threshold definitions;
- boundary policy;
- uncertainty model;
- graph topology and all adaptive-refinement events.

12.2 Suggested tabular event format

Each ABE should be stored as a row containing

$$\left[\text{run}, n, t, i, \mathbf{x}_i, h_i, R_i, R_i^{(u)}, R_i^{(p)}, R_i^{(c)}, \mathbf{u}_i, p_i, \delta_i, H_i \right]. \quad (12.1)$$

This table makes the boundary phenomenon analysable independently of the visual field data.

12.3 No hidden interpolation

Post-processing can inadvertently recreate the very continuum assumptions being investigated. Visualisation software commonly interpolates between cells and produces smooth surfaces. Therefore all figures should state whether interpolation has been used. At least one diagnostic view should display the native finite cells without smoothing.

Likewise, trajectory analyses should be performed on the actual recorded samples rather than on spline-interpolated time series unless interpolation is itself the subject of a declared comparison.

Chapter 13

Controls, Failure Modes and Falsification

13.1 The “alpha is only a grid size” objection

The strongest immediate objection is that the Alphonic limit is merely a chosen minimum mesh spacing. The study cannot answer this objection by terminology. It must demonstrate what is structurally different.

The distinction proposed here consists of three coupled rules: α is a hard admissibility constraint rather than a desired numerical resolution; prohibited refinements are logged as events rather than interpreted as missing approximations; and uncertainty/provenance are carried when the representation saturates.

Nevertheless, if all reported effects can be reproduced by an ordinary grid cutoff with no additional explanatory value, that is an important negative result. The research should state this openly.

13.2 Finite precision is not the same as finite ontology

Floating-point arithmetic is finite whether or not FSM is adopted. Therefore no result should be attributed to FSM merely because a computer has finite memory. The relevant experiment concerns declared finite spatial and symbolic admissibility, not accidental machine precision.

Runs should therefore be repeated at higher floating-point precision where feasible. If an apparent ABE phenomenon disappears when arithmetic precision changes while α remains fixed, the phenomenon is numerical rather than Alphonic.

13.3 Numerical instability

A CFL violation, unstable pressure solve or non-conservative flux may produce explosive values that resemble singular behaviour. The code should calculate conventional stability diagnostics and terminate or label invalid runs rather than classify them as FSM boundary

phenomena.

13.4 Threshold dependence

ABE definition depends initially on a threshold R_{crit} . The study must therefore sweep this threshold over a sensible range and determine whether qualitative results persist. A phenomenon appearing at only one arbitrary threshold is weak evidence.

13.5 Time-step dependence

If decreasing Δt changes the first ABE time dramatically while spatial parameters remain fixed, then the observed event may be a temporal discretisation artefact. A temporal convergence study is therefore mandatory before interpreting spatial Alphonic behaviour.

13.6 Mesh orientation and graph dependence

Vortex alignment with a Cartesian mesh can create artificial anisotropy. Selected cases should be repeated with shifted, rotated or irregular finite graphs. The primary conclusions should be based on features that survive these changes.

13.7 A finite graph can still blow up numerically

Perhaps the most important falsification condition is that $\alpha > 0$ does not guarantee bounded state amplitudes. A recurrence such as

$$x_{n+1} = x_n^2 \tag{13.1}$$

contains no spatial denominator tending to zero and can still grow without practical bound. Therefore the finite Navier–Stokes model must demonstrate boundedness through its conservation and interaction rules if boundedness is to be claimed.

If the finite fluid model develops unbounded or solver-divergent momentum while all spatial extents remain at least α , the simple proposition that the Alphonic limit “removes singularity” has failed. The result would refine the FSM hypothesis rather than invalidate the wider finite framework.

Chapter 14

Analysis Workflow

14.1 A complete reproducible study sequence

A researcher carrying out the programme can follow the sequence below.

- Step 1.** Implement the finite graph geometry, state records and provenance system without adaptive refinement.
- Step 2.** Validate mass and momentum accounting on static and simple transport tests.
- Step 3.** Implement the graph incompressibility projection and verify residual control.
- Step 4.** Implement viscous neighbour exchange and compare with known decaying smooth flows.
- Step 5.** Implement nonlinear finite advection with at least two flux schemes.
- Step 6.** Reproduce two-dimensional benchmark behaviour on fixed grids.
- Step 7.** Add adaptive refinement with a minimum permitted cell extent but do not yet label it Alphonic.
- Step 8.** Verify that refinement improves ordinary numerical agreement where expected.
- Step 9.** Activate the FSM rule: declare α , forbid child cells below it and log all rejected refinements.
- Step 10.** Implement the strict-refusal policy first, followed by uncertainty-expansion and conservative-redistribution policies.
- Step 11.** Run the three-dimensional Taylor–Green validation sweep.
- Step 12.** For each case, run a matched deeper-refined control from the same initial state.
- Step 13.** Store native finite fields, conservation diagnostics, ABE tables and complete manifests.
- Step 14.** Repeat selected runs under time-step variation, flux-rule variation and graph perturbation.

- Step 15.** Perform the α/L sweep and Reynolds-number sweep.
- Step 16.** Add the smooth high-shear initial-condition family.
- Step 17.** Introduce finite initial uncertainty and repeat a selected subset as ensembles.
- Step 18.** Perform delay reconstruction, recurrence analysis, perturbation-growth analysis and symbolic-complexity analysis on the resulting finite observables.
- Step 19.** Compare nonlinear trajectory changes with the time and location of ABE onset.
- Step 20.** Classify each result according to the interpretation scheme in Chapter 15.

14.2 Minimal first experiment

If computational resources are limited, the first complete proof-of-method need only contain one periodic three-dimensional Taylor–Green case, two values of α/L , one Reynolds number, one matched deeper-refined control, and one alternate flux scheme. The essential requirement is that the full admissibility and event-recording mechanism is present.

The research can then scale outward without changing its conceptual core.

Chapter 15

Discussion and Intended Contribution

15.1 What the study hopes to achieve

The most valuable outcome would not be a declaration that Navier–Stokes has been “solved”. The classical existence and smoothness problem is posed in a different mathematical framework and must be answered on its own terms. The present study instead hopes to make visible the representational work that occurs before that problem is formulated.

In the classical trajectory, a fluid is represented by fields over a continuum. Differentiability and smoothness then concern relations that remain meaningful under arbitrary refinement. The mathematical possibility of finite-time singularity asks whether those smooth relations can fail in three dimensions.

FSM changes the admissible domain of description. The finite graph contains no zero-extent spatial node and no operation that must be executed at $h = 0$. A large finite difference can still occur. Rapid nonlinear growth can still occur. Turbulent complexity can still occur. Numerical instability can still occur. What cannot occur by definition is an operational sequence of independently represented spatial distinctions passing below α .

This means that a classical singularity and an FSM boundary event should not be identified automatically. The research is designed to discover their relationship rather than assume it.

15.2 Possible outcome I: smooth finite saturation

The finite system may remain bounded while ABE events appear in regions where the deeper-refined control generates smaller-scale structure. Large-scale observables may remain close across the two runs even though the fine representations differ.

This would support the proposition that at least some continuum refinement demand can be re-expressed as finite representational saturation. It would not prove the non-existence of classical singularities. It would show that the physically executable finite model has a different endpoint than “gradient tends to infinity”.

15.3 Possible outcome II: uncertainty replaces refinement

The finite system may remain computationally stable but lose predictive distinction as uncertainty expands around unresolved regions. In that case the Alphonic boundary would not provide a more precise prediction. It would instead identify where the finite model ceases to justify additional detail.

This would be an important FSM result. A scientific representation need not answer every syntactically expressible question. The boundary of admissibility can itself be part of the answer.

15.4 Possible outcome III: finite-state instability remains

The model may exhibit rapidly growing or numerically unbounded state variables even though no cell is smaller than α . This would demonstrate that a finite spatial cutoff alone is insufficient to regularise the nonlinear dynamics.

Such a result would be theoretically valuable because it would separate two ideas that can otherwise be conflated: the inadmissibility of infinite spatial refinement and the boundedness of finite nonlinear state evolution.

15.5 Possible outcome IV: no distinctive FSM effect

The most conservative possibility is that, once implementation details are controlled, the Alphonic-boundary model behaves exactly like an ordinary finite-volume calculation stopped at a chosen grid scale, with no additional predictive or explanatory advantage.

That outcome would not erase the philosophical distinction between finite-first and continuum-first interpretation, but it would limit the claim that the distinction produces new practical fluid-dynamical consequences. This possibility must remain open for the project to function as research rather than demonstration by construction.

15.6 What would count as a strong result

A strong result would have several features simultaneously. ABE onset would be reproducible across transport schemes and mesh perturbations; it would correlate spatially with dynamically intense regions; its timing and density would scale systematically with α/L and Reynolds number; large-scale observables would remain controlled; nonlinear trajectory diagnostics would reveal a repeatable transition near boundary saturation; and uncertainty or finite redistribution would provide an interpretable replacement for otherwise continued refinement.

Even then, the correct conclusion would be restrained: the finite FSM representation exhibits a reproducible boundary phenomenon associated with the generation of sub-grid structure, and that phenomenon offers a different finite interpretation of the route toward continuum singularity-like behaviour.

15.7 Relation to classical mathematics

The project is potentially most useful when it does not position itself as an attack on classical analysis. Classical continuum mathematics can be regarded as an extraordinarily efficient extension of finite relations. It allows statements, proofs and asymptotic reasoning that no laboratory instrument performs directly. Its success is not diminished by identifying the assumptions that make it possible.

FSM simply asks that the extension not be silently identified with the measured system. In this view, the continuum Navier–Stokes equations and the finite FSM fluid graph become related descriptions with different admissibility rules.

That comparison may help separate three layers that are often compressed into one another:

$$\boxed{\text{physical fluid} \longrightarrow \text{finite registered fluid model} \longrightarrow \text{continuum mathematical model}}. \quad (15.1)$$

The research programme is primarily an investigation of the second arrow and of the consequences of reversing the usual explanatory priority between the last two terms.

15.8 Relation to Finite Symbolic Dynamics

The nonlinear analysis also provides a bridge from FSM to Finite Symbolic Dynamics. The flow simulation generates a finite trajectory

$$S^0 \rightarrow S^1 \rightarrow S^2 \rightarrow \dots \rightarrow S^N, \quad (15.2)$$

with each state carrying finite values, uncertainty and provenance. Delay reconstructions, recurrence relations and symbolic ordinal patterns can then be treated as secondary finite maps of that trajectory.

This is particularly useful because turbulence is often discussed through statistical averages or continuum spectra. An FSD analysis instead asks how the finite state path reorganises, bifurcates, recurs and loses or gains distinguishability as it approaches the Alphonic boundary.

The resulting object is not merely “a discretised PDE solution”. It is a finite symbolic dynamical history whose own admissibility conditions can be inspected.

15.9 The central hoped-for clarification

The central conceptual result can be stated carefully:

In a finite symbolic fluid model, a continuum singularity cannot be represented literally as an infinitely sharp spatial distinction. The corresponding observable question is whether the finite trajectory reaches a boundary at which its declared representation can no longer distinguish the structure demanded by its own refinement criteria.

This suggests a candidate FSM reformulation:

$$\boxed{\text{Singularity as infinity} \longrightarrow \text{boundary of admissible distinction}} \quad (15.3)$$

The arrow is deliberately not an equals sign. Establishing whether, when and how the two concepts correspond is the purpose of the proposed study.

Chapter 16

Recommended Deliverables

A complete first publication from the project should contain four linked artefacts.

- 1. The research paper or monograph.** This should present the finite formulation, the Alphonic rule, controls, results and interpretation.
- 2. The executable Python repository.** The repository should contain the solver, configuration files and analysis scripts. The smallest example should run on a modest machine even if high-resolution experiments require larger resources.
- 3. A finite provenance dataset.** Each published figure should be traceable to a run manifest and event log.
- 4. A visual atlas.** This should contain matched native-cell visualisations of FSM and deeper-refined controls, ABE maps, refinement histories, recurrence plots and reconstructed trajectories.

A useful later extension would be an interactive trajectory viewer in which the user can move through time, inspect a cell, read its state and provenance, and follow the chain of refinements that terminated at α .

Chapter 17

Conclusion

Navier–Stokes provides an unusually sharp test of Finite Symbolic Mechanics because the conventional theory is simultaneously practical, mathematically sophisticated and explicitly dependent upon a continuum field representation. The unresolved classical problem concerns the persistence of smoothness in that representation. FSM asks what becomes of the problem when the foundational description is constrained to finite, provenance-carrying distinctions whose spatial refinement stops at an Alphonic limit.

The research programme proposed here does not attempt to evade the nonlinear physics. Advection remains nonlinear. Three-dimensional vortex interactions remain complex. Conservation remains demanding. Pressure remains a coupled global constraint. Turbulence remains capable of creating structure over many scales. The proposed change lies elsewhere: the finite model never asks a physical or symbolic operation to cross from α to zero.

That gives the investigation a clean experimental form. Construct the finite system. Validate it. Allow refinement. Stop at α . Record every place where the ordinary refinement logic asks for more. Compare those places and times with deeper-refined controls. Carry uncertainty and provenance rather than smoothing the loss of distinction away. Then analyse the resulting histories not only as fields but as finite nonlinear trajectories.

If the programme succeeds, its contribution will not be the disappearance of Navier–Stokes difficulty. It will be a more precise account of where that difficulty resides. It may show that some questions conventionally expressed through infinity can be reformulated as questions about finite representational admissibility. It may show that other forms of nonlinear instability remain untouched by the Alphonic limit. Either result would sharpen FSM.

The strongest version of the project is therefore methodological rather than rhetorical. It turns the Alphonic limit into a rule that a program must obey, creates observable consequences when that rule becomes active, and makes those consequences available for direct comparison with established numerical practice. In doing so, it provides a route by which a foundational claim of FSM can be investigated through a recognisable problem of mathematical physics rather than left solely at the level of philosophical argument.

Appendix A

Suggested Initial Parameter File

```
experiment_name: taylor_green_fsm_alpha_sweep
model: incompressible_finite_graph

domain:
  type: periodic_cube
  length: 1.0

initial_condition:
  type: taylor_green
  U0: 1.0

physics:
  rho: 1.0
  reynolds_number: 400
  viscosity: derived_from_Re

fsm:
  alpha_over_L: 0.03125      # 1/32
  refinement_threshold: 2.0  # proposed; sweep later
  boundary_policy: strict_refusal
  carry_uncertainty: true
  carry_provenance: true

time:
  dt_policy: finite_stability_rule
  t_end: 10.0

numerics:
  base_grid: 8
  flux_scheme: conservative_upwind
  pressure_solver: graph_projection
  divergence_tolerance: 1.0e-8

recording:
  diagnostics_every_step: true
  field_checkpoint_interval: 0.05
  record_all_boundary_events: true
  native_cell_output: true
```

Listing A.1: Illustrative YAML-style experiment configuration.

The numerical values above are proposed starting values, not empirical constants. They should be altered during validation and sensitivity analysis.

Appendix B

Suggested Analysis Pseudocode

```
def analyse_run(run):
    series = extract_observables(
        run,
        names=[
            "max_finite_gradient",
            "abe_count",
            "abe_fraction",
            "circulation_norm",
            "uncertainty_spread",
            "quadratic_velocity_norm"
        ]
    )

    first_abe = first_time(series["abe_count"] > 0)

    for name, s in series.items():
        delay = choose_delay_by_mutual_information(s)
        dimension = choose_embedding_dimension_fnn(s, delay)

        Y = delay_embed(s, delay, dimension)

        recurrence = recurrence_matrix(
            Y,
            epsilon=choose_finite_recurrence_threshold(Y)
        )

        rqa = recurrence_quantification(recurrence)
        perm_entropy = permutation_entropy(s, delay=delay)

        store_analysis(name, delay, dimension, rqa, perm_entropy)

    if run.has_perturbation_ensemble:
        lyap_like = finite_time_perturbation_growth(run.ensemble)
        store_analysis("finite_time_growth", lyap_like)

    compare_pre_and_post_boundary(first_abe)
    return analysis_report()
```

Listing B.1: Nonlinear analysis workflow.

Appendix C

Glossary

Alphonic limit α

Proposed FSM lower bound on independently admissible spatial distinction within the finite representation.

Alphonic boundary event (ABE)

A recorded event in which a declared refinement criterion is exceeded but the required child representation would violate α .

Admissibility

The declared conditions under which a finite symbol, relation or update is accepted as part of the model.

Finite Symbolic Mechanics (FSM)

The proposed framework in which scientific representations are treated as finite symbolic relations with finite distinction, uncertainty, provenance and explicit admissibility conditions.

Finite Symbolic Dynamics (FSD)

Dynamical analysis of finite symbolic trajectories and their transitions, recurrence and organisation.

Finite graph fluid

A fluid representation in which finite cells are nodes and admissible transfers between neighbouring cells are edges or face relations.

Continuum compression

The interpretation of a smooth mathematical field as a compact higher-order representation derived from finite relations, rather than as the foundational object from which finite measurements are merely approximations.

Provenance

The retained history of how a finite state, symbol or reported quantity was produced.

Refinement demand

A finite diagnostic indicating that the current representation fails a declared local variation, conservation or uncertainty criterion and would conventionally be refined.

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